1) If $z = \frac{xy}{x^2+y^2}$, find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$

Ans

$$\frac{\partial z}{\partial x} = \frac{(x^2+y^2) \cdot y - (xy) \cdot 2x}{(x^2+y^2)}$$

$$\frac{\partial z}{\partial y} = \frac{(x^2+y^2) \cdot x - (xy) \cdot 2y}{(x^2+y^2)}$$

2 Show that the equation $u(x,t) = \sin(x - ct)$. Satisfies wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Ans.

$$\frac{\partial u}{\partial t} = \cos(x-ct) \cdot -c$$

$$\frac{\partial^2 u}{\partial t^2} = -\sin(x-ct) \cdot -c \cdot -c$$

$$= -c^2 \sin(x-ct)$$

$$\frac{\partial u}{\partial x} = \cos(x-ct)$$

$$\frac{\partial^2 u}{\partial x^2} = -\sin(x-ct)$$

$$\therefore \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$= -c^2 \sin(x-ct) = c^2 \cdot -\sin(x-ct)$$

$$= -c^2 \sin(x-ct)$$

3a) If $w = F(P, Q, R)$ where $P = x - y$, $Q = y - z$, $R = z - x$ Prove that

$$\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Ans a)

Let $P = x - y$

$Q = y - z$

$R = z - x$

$$\begin{array}{ccc} & w & \\ P & Q & R \\ x, y & y\ z & z\ x \end{array}$$

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial P} \cdot \frac{\partial w}{\partial x} + \frac{\partial w}{\partial R} \frac{\partial R}{\partial x}$$

$$= \frac{\partial w}{\partial P} \cdot 1 + \frac{\partial w}{\partial R} \cdot -1$$

$$= \frac{\partial w}{\partial P} - \frac{\partial w}{\partial R}$$

$$\frac{\partial w}{\partial y} = \frac{\partial w}{\partial P} \frac{\partial Q}{\partial y} + \frac{\partial w}{\partial Q} \frac{\partial Q}{\partial y}$$

$$= \frac{\partial w}{\partial P} \cdot -1 + \frac{\partial w}{\partial Q} \cdot 1$$

$$= -\frac{\partial w}{\partial P} + \frac{\partial w}{\partial Q}$$

$$\frac{\partial w}{\partial z} = \frac{\partial w}{\partial Q} \cdot \frac{\partial Q}{\partial z} + \frac{\partial w}{\partial R} \frac{\partial R}{\partial z}$$

$$= \frac{\partial w}{\partial Q} \cdot -1 + \frac{\partial w}{\partial R} \cdot 1$$

$$= -\frac{\partial w}{\partial Q} + \frac{\partial w}{\partial R}$$

$$\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z} = 0$$

$$\therefore \frac{\partial w}{\partial P} - \frac{\partial w}{\partial R} + -\frac{\partial w}{\partial P} + \frac{\partial w}{\partial Q} + -\frac{\partial w}{\partial Q} + \frac{\partial w}{\partial R} = \underline{\underline{0}}$$

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3) b Locate all relative extrema and saddle points of $f(x,y) = 4xy - x^4 - y^2$

Ans) $F(x,y) = 4xy - x^4 - y^2$

$f_x = 4y - 4x^3 = 0 \quad -(1)$

$f_y = 4x - 4y^3 = 0 \quad -(2)$

$4y - 4x^3 = 0$

$4y = 4x^3$

$y = x^3$

Subsitute $y = x^3$ in (2)

$4x - 4y^3 = 0$

$4x - 4(x^3)^3 = 0$

$4x - 4x^9 = 0$

$x - x^9 = 0$

$x(1 - x^8) = 0$

$x = 0, x^8 = 1$

$x = 0, x = \pm 1$

When $x = 0$

$4y - 4x^3 = 0$

$4y - 4 \times 0 = 0$

$4y = 0$

$y = 0$

When $x = 1$

$4y - 4x^3 = 0$

$4y - 4 \times 1^3 = 0$

$4y = 4$

$y = 1$

When $x = -1$

$4y - 4 \times (-1)^3 = 0$

$4y + 4 = 0$

$4y = -4$

$y = -1$

$f_{xx} = -4 \cdot 3x^2$
$= -12x^2$

$f_{yy} = -12y^2$

$f_{xy} = 4$

$(0,0), (1,1), (-1,1)$ are critical points

Critical points	f_{xx}	f_{yy}	f_{xy}	$D = f_{xx}f_{yy} - (f_{xy})^2$
(0,0)	0	0	4	-16
(1,1)	-12	-12	4	128
(-1,-1)	12	12	4	128

At (0,0) $D = -16 < 0$ So funtion has saddle point

At (1,1) $D = 128 > 0$, $f_{xx} < 0$ funtion has relative maximum

At (-1,-1) $D = 128 > 0$, $f_{xx} > 0$ funtion has relative minimum